



AIR CAIRO CARGO · WEBSITE, AGENT PORTAL & MOBILE APP

Hosting, Infrastructure & Operations Proposal

Two-continent Azure hosting, disaster recovery, 24/7 monitoring and on-call support — including high-level design, delivery milestones and commercial schedule.

99.9%

AVAILABILITY TARGET

4 weeks

SETUP · 2 SPRINTS

≤15 min

RECOVERY POINT

≤4 hr

RECOVERY TIME

A resilient, cost-efficient hosting platform for Air Cairo Cargo

We propose a secure, horizontally scalable hosting platform on Microsoft Azure across **two sites on two continents**, as requested in RFP §2.4 – covering all environments, edge security, disaster recovery, 24/7 monitoring and on-call support under a dedicated Hosting SLA.

Each application service runs on its own group of small virtual machines that scales out automatically under load. Data runs on Azure SQL Database (Microsoft SQL Server engine), with managed queueing and caching. The disaster recovery site runs at minimum capacity and scales up automatically when a failover happens – keeping the running cost low while meeting the RFP recovery requirements. The full setup is delivered in **two sprints (4 weeks)**.

99.9% MONTHLY AVAILABILITY	≤ 15 min RECOVERY POINT (RPO)	≤ 4 hr RECOVERY TIME (RTO)	2 continents UAE NORTH · ITALY NORTH
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Why this design

SCALE OUT, NOT UP Horizontal by design Small VMs per service, 2 → 3 nodes automatically. Any node can be added or removed at any time.	PAY FOR WHAT RUNS Minimum-capacity DR The DR site stays small and grows to production size only during a real failover.	TRANSPARENT BILLING Two separate bills Our services are invoiced by us. Cloud consumption is paid by the client directly – no resale, no mark-up.
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At a glance

Item	Summary	Price
Infrastructure setup	2 sprints · 4 weeks · 160 hours · 50% per accepted sprint	\$8,000 fixed
On-call support	24/7 monitoring, on-call availability, monthly operations	from \$500 / mo
Cloud consumption	Azure + Cloudflare, paid by the client directly	~\$1.7K–2.7K / mo
Combined year one	Setup + support + cloud (estimate)	\$34K–55K

Contents

01	Executive summary	Overview and key figures
02	Scope of services	In scope, out of scope, shared responsibilities
03	High-level design	Architecture, regions, HA/DR, security, operations
04	Environments	Dev, QA, UAT, Staging, Production, DR
05	Requirements traceability	RFP infrastructure requirements mapped
06	Hosting SLA & support model	Severities, terms, escalation
07	Delivery milestones	2 sprints, deliverables, acceptance, payment
08	Commercial schedule	Bill A (our services) · Bill B (cloud)
09	Assumptions & clarifications	Basis of this proposal
A	Appendix – Architecture board	One-page HLD overview (A3)

What we deliver – and what we don't

Our scope is hosting, infrastructure, operations and the Hosting SLA. Application design, development and integrations are delivered by the application vendor.

IN SCOPE

- Hosting design and build on Azure – Site A primary, Site B DR
- Environments: Dev, QA, UAT, Staging, Production, DR
- Network security, WAF, DDoS and bot mitigation
- Web server management: configuration, tuning, patching
- Azure SQL setup, geo-replication, backup retention, performance monitoring
- Managed queue (Service Bus) and cache (Cosmos DB)
- Backup & restore, DR plan and runbook, one DR failover test
- CI/CD pipelines, source code repository and versioning
- 24/7 automated monitoring, alerting and self-healing
- Secure remote access for support
- On-call support under a separate Hosting SLA
- Documents: HLD, Backup Policy, DR Plan, Data Breach Policy, SLA monitoring method, Escalation Matrix

OUT OF SCOPE

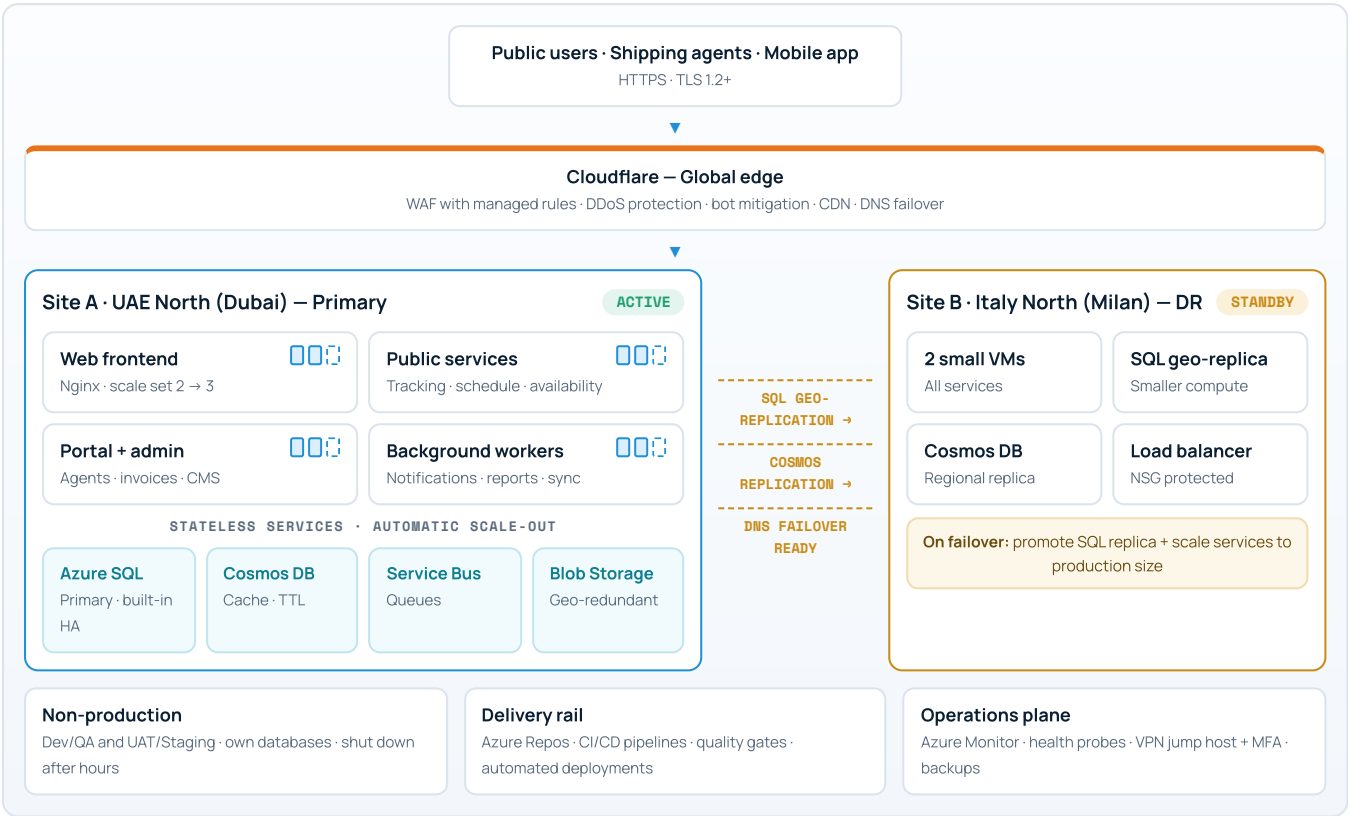
- Application design and development – website, agent portal, admin/CMS, mobile app
- Integrations – Cargo Flash, SAP, payment gateway, SMS, maps, chatbot
- Enterprise DXP licensing and implementation
- Application bug fixing and content management support
- Load test execution, go-live support and hypercare – available at on-call hourly rates

Cloud consumption (Azure, Cloudflare) is billed to the client directly and is not part of our fees.

Shared responsibilities with the application vendor

Item	Our part	Application vendor's part	RFP
Horizontal scaling	Scale sets, load balancing, autoscale rules	Stateless services – no local session or files	§2.4
Self-healing	Health probes, automatic restart, node replacement	Health-check endpoint in each service	§2.2.1
Caching	Cosmos DB cache store, CDN	Cache logic and TTLs in the code	§2.2.9
Messaging	Service Bus namespace, queues, access	Producers and consumers in the code	§2.3.19
Source repository	Platform, access control, backups	Code, branching, reviews	§2.2.8
Load testing	Test tooling and infrastructure scaling	Test scenarios and results analysis	§2.2.5
Security testing	Infrastructure hardening and scans	Application security fixes	§2.2.5
Build optimization	Minification and image steps in the pipeline	Build configuration	§2.2.9
Encryption at rest	Disk encryption, SQL TDE	Field-level encryption if needed	§2.3.18
Audit & anomalies	Log platform and alerts	Application audit events	§2.2.4
Mobile release	Build and release pipeline	App code; store accounts owned by the client	§2.2.3

Two sites, two continents, one operating model



Regions

Site	Azure region	Continent	Role
Site A	UAE North (Dubai)	Asia	Primary, active
Site B	Italy North (Milan) — alternative: West Europe	Europe	DR, minimum capacity, scales up on failover

Egypt has no Azure region; both regions give low latency to Egypt. Data residency to be confirmed by the client (Law 151/2020).

Components

Layer	Components	Scaling
Edge	Cloudflare: WAF, DDoS, bot protection, TLS, CDN, DNS failover	Managed
Network	Load balancers, Network Security Groups, private subnets	—
Presentation	Web frontend (Nginx)	Horizontal, 2 → 3 nodes
Services	.NET Core / Java microservices, grouped per service	Horizontal, 2 → 3 per group
Workers	Background jobs — notifications, reports, sync	Horizontal, 2 → 3 nodes
Data	Azure SQL Database (SQL Server engine)	Managed, built-in HA
Cache	Azure Cosmos DB (key-value, TTL)	Managed, autoscale
Messaging	Azure Service Bus	Managed
Files	Azure Blob Storage — documents, media, download center	Managed, geo-redundant
Management	Azure Monitor, log analytics, alerting, jump host with VPN	—

The four service groups are a starting assumption; final grouping follows the application vendor's microservices design, each group with its own scale set.

Resilience, security and operations

High availability & disaster recovery

Item	Proposed value
Availability target	99.9% monthly (production, excluding planned maintenance)
Site A	2+ nodes per service across availability zones; managed SQL with built-in HA
DR model	Minimum-capacity standby in Site B
DR database	Geo-replica on a smaller compute size, scaled up to production size at failover
DR services	2 small VMs running all services; scale sets expand to production size at failover
Failover	DNS failover at the edge + geo-replica promotion + automatic scale-out
RPO / RTO	≤ 15 minutes / ≤ 4 hours (aligned with the Critical resolve time)
DR test	One failover test during setup; then twice a year

Backup policy

Data	Method	Retention
Azure SQL	Automated full, differential and log backups (point-in-time restore)	14 days
Azure SQL long-term	Weekly and monthly long-term retention	4 weeks weekly · 12 months monthly
Blob storage	Geo-redundant storage + soft delete	30 days soft delete
Server configuration	Documented configuration; VM images versioned	All versions
Restore test	Monthly on production backups	Report to client

SECURITY

Protected by default

- TLS 1.2+ everywhere; certificates auto-renewed
- Disk encryption on all servers; SQL TDE for customer data
- No public server access; admin via jump host with VPN + MFA
- Monthly OS and web server patching; emergency patching for critical vulnerabilities
- Central security logging and real-time anomaly alerts
- Data Breach Policy: detection, containment, notification, recovery, review

OPERATIONS

Observable & self-healing

- 24/7 automated uptime checks – website, portal, APIs
- Server, web, service, database, cache and queue metrics
- Alerts routed to the on-call engineer by phone and app
- Automatic restart of failed services and replacement of unhealthy nodes
- Scale-out on load
- Monthly availability and performance report

Exit strategy

Documented setup

Every server, network and service configuration is documented and handed over.

Standard runtime

Linux VMs with Nginx – portable to any cloud or on premises.

Standard data

SQL Server engine – exportable as standard backups at any time.

Replaceable services

Cache and queue used through standard patterns.

Every stage of the release path

Each environment has its own database, configuration and pipeline stage: **Dev → QA → UAT → Staging → Production**.

Environment	Purpose	RFP	Setup
Development	Developer integration and daily builds	§2.2.8	Dev/QA servers (2 small VMs) · own database · shut down after hours
QA / Test	Functional and integration testing	§2.2.8 · §2.4	Same Dev/QA servers · separate database
UAT	Client acceptance testing	§2.2.8	UAT/Staging servers (2 small VMs) · own database
Staging	Production-like pre-release validation	§2.4	Same UAT/Staging servers · separate database · production configuration
Production	Live service — Site A	§2.4	Scale set per service, 2 → 3 nodes
DR	Standby — Site B	§2.4	Minimum capacity, scales up on failover

RFP infrastructure requirements, mapped

ID	RFP requirement	Ref	Our response
H-01	Scalable, efficient hosting with HLD	§2.4	Scale set per service, horizontal autoscale 2 → 3
H-02	High availability (redundancy)	§2.4	2+ nodes per service across zones; managed SQL with built-in HA
H-03	Network firewall	§2.4	Stateful Network Security Groups + host firewalls; optional Azure Firewall Basic
H-04	Web application firewall	§2.4	Edge WAF with managed rule sets (Cloudflare)
H-05	DDoS prevention and bot mitigation	§2.4	Cloudflare DDoS and bot protection + Azure basic DDoS protection
H-06	Two sites, two geo locations, preferably two continents	§2.4	Site A: UAE North (Asia) · Site B: Italy North (Europe)
H-07	Minimum bandwidth 3TB per site	§2.4	Capacity provided at both sites
H-08	Staging environment	§2.4	Dedicated UAT/Staging environment
H-09	Test environment by vendor	§2.4	Dedicated Dev/QA environment
H-10	24×365 availability, planned outages notified	§2.4	99.9% monthly target; maintenance windows notified in advance
H-11	Reliable backups at a safe location	§2.4	Automated SQL backups, geo-redundant storage, long-term retention
H-12	DR implemented + backup policy	§2.4	Scale-up DR site in Site B; Backup Policy document
O-01	Web server config, tuning, patching, uptime monitoring	§2.2.6	Nginx managed by us
O-02	Secure remote support; email, phone, chat; proactive monitoring	§2.2.7	Jump host with VPN; support channels; automated alerting
O-03	Deployments, backup/restore, health, pipelines, upgrades, DB patching	§2.3.20	Operations runbooks; DB patching by the managed service
O-04	Self-healing	§2.2.1	Health probes, auto-restart, auto-replace of failed nodes
S-01	HTTPS/TLS for all data exchange	§2.2.4	TLS 1.2+ end to end
S-02	Encryption at rest for customer data	§2.3.18	Disk encryption + SQL TDE
S-03	Real-time anomaly monitoring	§2.3.18	Central logs, security alerts
S-04	Data breach policy	§2.3.18	Delivered as a document
A-01	Three-tier architecture	§2.3.19	Presentation, service and data tiers on separate subnets
A-02	Microsoft SQL Server	§2.3.19	Azure SQL Database (SQL Server engine)
A-03	Azure compatible, scalable, secure	§2.3.19	Hosted on Azure
A-04	Inter-service messaging (message queues)	§2.3.19	Azure Service Bus
L-01	Separate Hosting SLA	§2.2.8	Section 06
L-02	Dev, QA, UAT servers	§2.2.8	Section 04
L-03	24/7 uptime monitoring	§2.2.8	Section 03
L-04	Escalation matrix, man-day rate	§2.2.8	Sections 06 and 08

On-call, ticket-based support – 24/7 for what matters

Severity levels RFP #2.2.8

Severity	Definition	Response	Resolve	Coverage
CRITICAL	Main website or service down; impact on revenue or financial transactions	1 hour	4 hours	24×7 incl. holidays
MEDIUM	Decreased performance; low financial, regulatory or customer impact	2 hours	1 business day	5×8
MINOR	Minor degradation, no financial or customer impact	4 hours	2 business days	5×8
CHANGE	Change request, as agreed per request	Per estimate	Per estimate	5×8

Business hours: Sunday to Thursday, Egypt time.

SUPPORT MODEL

How incidents are handled

- **24/7 automated monitoring** detects issues and alerts the on-call engineer
- **On-call engineers** cover Critical incidents 24×7 within the 1-hour response
- **All other requests** via tickets during business hours
- **Routine operations** on a fixed monthly schedule
- **Channels:** ticket portal, email, phone, chat

SLA TERMS

Proposed Hosting SLA

- Availability: 99.9% monthly for production
- Planned maintenance notified ≥ 5 business days ahead, off-peak
- Service credits apply to hosting availability only
- Excluded: application code, Cargo Flash, SAP, payment gateway, Azure platform outages, third parties, client-side changes
- Monthly SLA report · quarterly SLA review

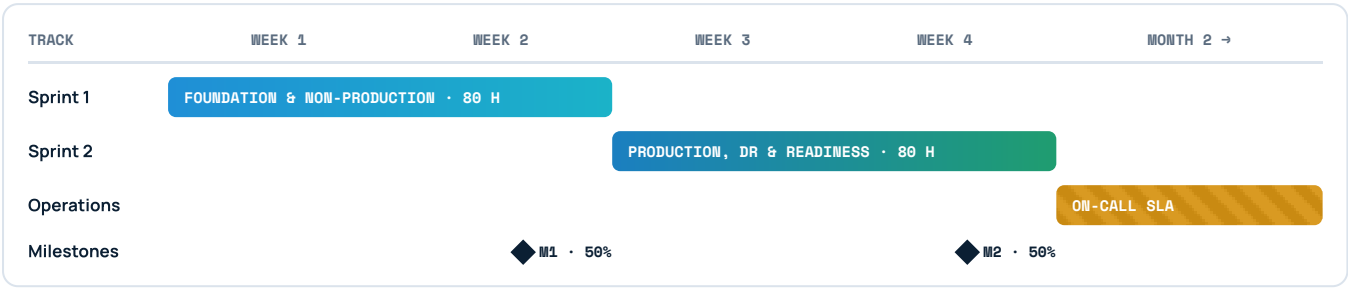
Escalation matrix

Level	Role	When
L1	On-call engineer	All incidents
L2	Senior infrastructure engineer	Not resolved in 1 hr (Critical)
L3	Lead architect	Not resolved in 2 hrs (Critical)
L4	Account director	SLA breach risk

Routine operations calendar

Activity	Frequency
OS and web server patching	Monthly
Backup verification and restore test	Monthly
Performance, scaling and cost review	Monthly
Availability and SLA report	Monthly
Security review and vulnerability scan	Quarterly
DR failover test	Twice a year

Two sprints · four weeks · 160 hours



M1 · Sprint 1 — Foundation & Non-Production		WEEKS 1-2 · 80 HOURS · 50% · \$4,000
Deliverable	Detail	
Discovery & design	Requirements confirmation, Hosting HLD and LLD	
Azure foundation	Subscriptions, networking in both sites, NSGs, jump host with VPN	
Edge security	Cloudflare WAF, DDoS, bot protection, TLS, DNS	
CI/CD & repository	Source repository; pipeline stages Dev → QA → UAT → Staging → Production	
Dev/QA environment	Servers, databases, cache, queue, deployment	
UAT/Staging environment	Servers, databases, cache, queue, deployment	
Setup documentation	Configuration of all environments documented for handover	
Acceptance ✓ Dev/QA and UAT/Staging environments running; HLD approved.		

M2 · Sprint 2 — Production, DR & Operations Readiness		WEEKS 3-4 · 80 HOURS · 50% · \$4,000
Deliverable	Detail	
Production · Site A	Scale sets per service (2 → 3), Azure SQL, Cosmos DB, Service Bus, Blob	
DR · Site B	Minimum-capacity services, SQL geo-replica, scale-up automation	
Monitoring & self-healing	Uptime checks, metrics, alerts, on-call routing, auto-restart and node replacement	
Security hardening	Encryption, access control, patch baseline, security logging	
Backup & DR	Backup Policy, DR Plan, DR runbook, one failover test	
Documents	Data Breach Policy, SLA monitoring method, Escalation Matrix	
Handover	Runbooks and knowledge transfer to client staff	
Acceptance ✓ Production and DR ready; DR failover test passed; documents delivered.		

Go-live date depends on the application vendor's schedule. Go-live support, load test execution and hypercare are billed at on-call hourly rates.

Two bills – full transparency

Bill A is invoiced by us. **Bill B** is paid by the client directly to Microsoft and Cloudflare – we do not resell or mark up cloud consumption. The Azure subscription and Cloudflare account are owned by the client; we operate them with delegated access.

Bill A · Our services		USD · INVOICED BY US
Item	Price	
Infrastructure setup · 2 sprints · 160 hrs × \$50	\$8,000 fixed	
↳ Sprint 1 acceptance (M1) · 50%	\$4,000	
↳ Sprint 2 acceptance (M2) · 50%	\$4,000	
Monthly minimum · includes 10 business hours (24/7 monitoring, on-call availability, monthly patching, backup checks, monthly report)	\$500 / mo	
Business hours · Sunday–Thursday, Egypt time	\$50 / hr	
After hours, weekends and holidays (24/7)	\$75 / hr	
Critical call-out	2 hrs minimum	
Man-day · business hours	\$400 / day	
Year one · setup + support (estimate)	\$14K–\$23K	

Bill B · Cloud consumption

USD / MONTH · PAID BY THE CLIENT DIRECTLY

Azure environments	\$ / month	Shared & totals	USD
Site A Production · 8–12 VMs + managed	1,040–1,340	Cloudflare Pro to Business	25–250 / mo
↳ Service VMs · 4 groups × 2–3	504–756	Outbound traffic (egress)	20–260 / mo
↳ Azure SQL primary · General Purpose	~350	Azure DevOps hosted build agents	~40 / mo
↳ Cosmos, Service Bus, Blob, LB, logs, jump, disks	185–235	SQL long-term backup retention	~20 / mo
Site B DR · 2 VMs + managed replicas	300–410	Optional · Azure Firewall Basic, per site	~290 / mo
↳ DR service VMs	~140	DR scale-up · pro-rata, failover days only	+800–1K / mo
↳ SQL geo-replica · smaller compute	120–180	Normal operation · PAYG	1.7K–2.7K / mo
↳ Cosmos, load balancer, disks	40–90	PAYG yearly	20K–32K / yr
Dev / QA environment	110–160	1-year reserved production VMs	1.55K–2.5K / mo
UAT / Staging environment	140–190	Reserved yearly	19K–30K / yr
Non-production cache + queue	~10	Footprint · all environments	14–18 small VMs

\$8,000 ONE-TIME SETUP	~\$2,200 MONTHLY MINIMUM · A + B	\$14K–23K BILL A · YEAR ONE	\$34K–55K COMBINED YEAR ONE
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Region note. Tables use Italy North prices; UAE North is ~10–15% higher. With the proposed layout (primary in UAE North, DR in Italy North) expect roughly +\$100–200 per month over the table figures.

These figures are rough estimates, not quotations. Actual costs depend on the final regions, usage, traffic and Microsoft pricing at purchase time, and will be confirmed with the Azure Pricing Calculator. Sizing is an initial baseline; components and sizes may be adjusted during hosting based on actual usage and monitoring. Changes that affect cost are agreed with the client beforehand.

Basis of this proposal

Assumptions

01	Application services are stateless , run on Linux, and expose a health-check endpoint each.
02	Application services use Azure SQL, Cosmos DB and Service Bus through standard SDKs.
03	The Azure subscription and Cloudflare account are owned and paid by the client directly; we operate them with delegated access and do not resell cloud services.
04	Domain, DNS access and SSL ownership are provided by the client.
05	Mobile store accounts (Apple, Google) are owned by the client.
06	Setup scope is fixed to Section 07; additional work is billed at on-call hourly rates.
07	Unused monthly minimum hours do not roll over.
08	Prices are valid for 30 days.

Clarifications requested

Q1	Is a minimum-capacity DR site that scales up on failover acceptable?
Q2	3TB bandwidth: per month? Per environment or production only?
Q3	Target availability %, RPO and RTO expected by the client?
Q4	Data residency: is hosting outside Egypt accepted?
Q5	Is a dedicated firewall appliance required, or are Network Security Groups accepted?
Q6	Expected users — public visitors, agents, admins — and peak concurrent users?
Q7	Expected data volume and AWBs per month?
Q8	Hosting SLA duration: 12 months?
Q9	Confirm the client will own and pay the Azure subscription and Cloudflare account directly.
Q10	Allowed maintenance window per month?
Q11	Is there an existing SOC/SIEM to integrate with?
Q12	Bidder instructions, deadline, evaluation criteria and commercial terms; are partial bids or consortiums allowed?

Acceptance

For Air Cairo Cargo — name, title, signature, date

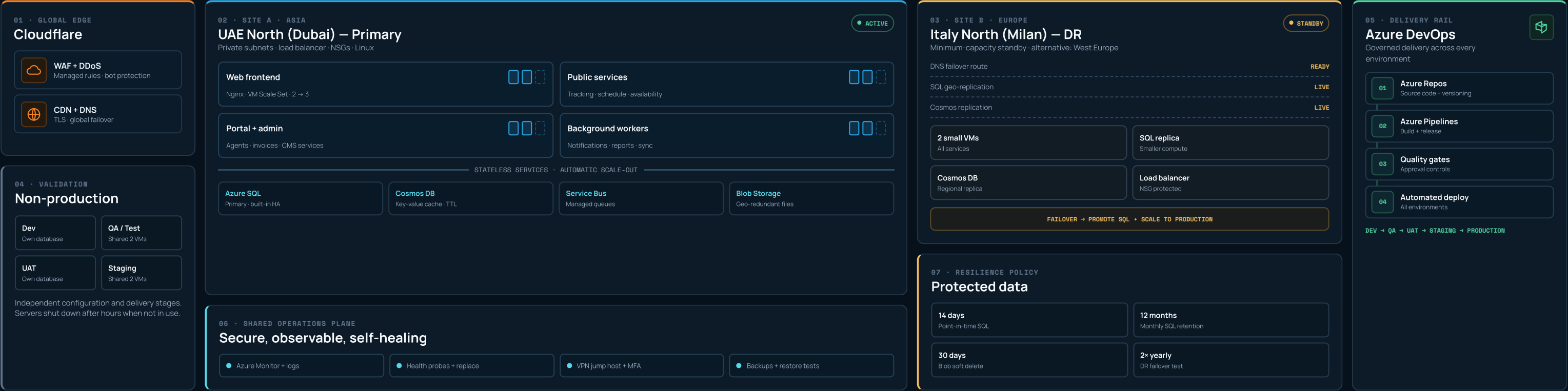
For the vendor — name, title, signature, date

EXECUTIVE HIGH-LEVEL DESIGN · TWO CONTINENTS

Resilient Azure hosting for cargo operations

Secure edge delivery, horizontally scaled Linux services, managed data, and a minimum-capacity disaster recovery site—operated under a dedicated hosting SLA.

● LIVE REQUEST PATH



05 · DELIVERY RAIL

Azure DevOps

Governed delivery across every environment

01

Azure Repos

Source code + versioning

02

Azure Pipelines

Build + release

03

Quality gates

Approval controls

04

Automated deploy

All environments

DEV → QA → UAT → STAGING → PRODUCTION

\$8,000 Infrastructure setup · fixed 2 sprints · 4 weeks · 160 hours · 50% per accepted sprint	\$1.7K–\$2.7K Azure + Cloudflare · monthly Paid directly by the client. Approximate PAYG Linux list prices.	\$500+ On-call support · monthly Minimum covers 10 business hours; \$50/hr business, \$75/hr after hours.
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08 · COMPLETE COMMERCIAL SCHEDULE			
All proposal costs			
Bill A is invoiced by us. Bill B is paid by the client directly to Microsoft and Cloudflare; we do not resell or mark up cloud consumption.			
Bill A · Our services		USD	
Infrastructure setup · 160 hrs × \$50		\$8,000 fixed	
Sprint 1 acceptance · 50%		\$4,000	
Sprint 2 acceptance · 50%		\$4,000	
Monthly minimum · includes 10 business hrs		\$500 / mo	
Business hours · Sun–Thu, Egypt time		\$50 / hr	
After hours, weekends and holidays		\$75 / hr	
Critical call-out		2 hrs min	
Mon-day · business hours		\$400 / day	
Year one · setup + support		\$14K–\$23K	
Bill B · Azure environments		USD / month	
Site A Production · 8–12 VMs + managed services		\$1,040–\$1,340	
Site A service VMs · 4 groups × 2–3		\$504–\$756	
Azure SQL primary · General Purpose		~\$350	
Cosmos DB / Service Bus / Blob / LB / logs / jump / disks		\$185–\$235	
Site B DR · 2 VMs + managed replicas		\$300–\$410	
DR service VMs		~\$140	
DR SQL geo-replica · smaller compute		\$120–\$180	
DR Cosmos / load balancer / disks		\$40–\$90	
Dev / QA environment		\$110–\$160	
UAT / Staging environment		\$140–\$190	
Non-production cache + queue		~\$10	
Bill B · Shared, totals & variables		USD	
Cloudflare Pro to Business		\$25–\$250 / mo	
Outbound traffic / egress		\$20–\$260 / mo	
Azure DevOps hosted build agents		~\$40 / mo	
SQL long-term backup retention		~\$20 / mo	
Optional Azure Firewall Basic · per site		~\$290 / mo	
DR active scale-up · pro-rata for failover days		+\$800–\$1K / mo	
Normal operation · PAYG		\$1.7K–\$2.7K / mo	
Normal operation · PAYG yearly		\$20K–\$32K / yr	
1-year reserved production VMs		\$1.55K–\$2.5K / mo	
Reserved scenario yearly		\$19K–\$30K / yr	
Total footprint · all environments		14–18 small VMs	
ONE-TIME SETUP	MONTHLY MINIMUM · SERVICES + CLOUD	BILL A · YEAR ONE	COMBINED YEAR ONE
\$8,000	~\$2,200	\$14K–\$23K	\$34K–\$55K
TABLES USE ITALY NORTH PRICES · UAE NORTH IS ~10–15% HIGHER	PROPOSED · A = UAE ~\$1,150–1,540 · B = ITALY ~\$300–410	REVERSE · A = ITALY ~\$1,040–1,340 · B = UAE ~\$330–470	UAE PRIMARY VS ITALY PRIMARY · SAME RATIO ON DR FAILOVER
Region note	+~\$100–200 / mo	+~\$30–60 / mo	+~\$70–140 / mo

These figures are rough estimates, not accurate quotations. Actual costs depend on the final regions, usage, traffic and Microsoft pricing at purchase time, and will be confirmed with the Azure Pricing Calculator. This sizing is an initial baseline: features, components and sizes may be added or adjusted during hosting based on actual usage, performance and monitoring results. Changes that affect cost are agreed with the client beforehand.

Scope: hosting, infrastructure, operations and SLA. Application development and integrations are excluded.

Final figures will be confirmed with the Azure Pricing Calculator.